

Total No. of Questions : 4]

SEAT No. :

PE559

[Total No. of Pages : 2

[6578]-32

S.E. (Artificial Intelligence and Machine Learning) (Insem)

DISCRETE MATHEMATICS

(2019 Pattern) (Semester - III) (218541)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if Necessary.

Q1) a) In the class, 20 students have Lenovo laptops, 15 students have Sony laptops and 8 students have both Lenovo and Sony laptops. How many students are there in the class? **[4]**

b) Construct a truth table for each of these compound propositions. **[5]**

i) $(p \vee q) \rightarrow q$

ii) $(p \vee q) \rightarrow (p \wedge q)$

c) Use mathematical induction to show that : **[6]**
 $n^3 + 2n$ is divisible by 3 for all $n \geq 1$

OR

Q2) a) How many different license plates are available if each plate contains a sequence of 3 letters followed by 4 even digits (0, 2, 4, 6, 8) where 0 cannot be taken as 1st digit? **[5]**

b) $A = \{x : x \text{ is a natural number less than or equal to 10 and divisible by 2}\}$
and $B = \{x : x \text{ is an odd natural number less than or equal to 10}\}$ **[5]**

Find i) $A - B$

ii) $B - A$

iii) is $A - B = B - A$?

c) What is CNF? Convert $p \leftrightarrow q$ into CNF. **[5]**

P.T.O.

- Q3)** a) In a country club, 60% of women play tennis, 40% play golf, and 20% play both. A woman is chosen at random. Given that she plays tennis, what is the probability that she plays golf? [5]
- b) A committee of 5 people is to be formed from a group of 4 men and 7 women. How many possible committees can be formed if at least 3 women are on the committee? [5]
- c) A man is informed that when a pair of dice were rolled, the result was seven. How much information is there in this message? [5]

OR

- Q4)** a) A box contains 6 white balls and 5 black balls find the number of ways 4 balls can be drawn from the box if [6]
- i) Two must be white
- ii) All of them must have same color
- b) Consider the experiment of tossing a coin three times. What is the probability of getting exactly two tails? [4]
- c) In how many ways can the letters of the word PERMUTATIONS be arranged if the vowels are all together? [5]

